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[5252]-104

S.E. Civil (First Semester) EXAMINATION, 2017

SURVEYING

(2012 PATTERN)

Time : Two Hours

Maximum Marks : 50

1. (a) Enlist the methods of plane table surveying. Give *two* limitations of each method. [6]
- (b) Following readings were observed during a reciprocal leveling with one level. [6]

Instrument at	Staff Readings on		Remark
	P	Q	
P	0.656	2.097	Distance between A and B is 950 m
Q	0.867	2.298	

Find :

- (a) the true R.L. of B, if R.L. of A = 378.650 m.
- (b) the combined correction due to curvature and refraction.
- (c) the error in the collimation adjustment of the instrument.

Or

2. (a) Find the included angles of the closed traverse PQRSTP and correct them for the corrections, if any. [7]

Line	PQ	QR	RS	ST	TP
F.B.	242°00'	89°45'	70°00'	292°45'	20°00'
B.B.	63°00'	270°15'	250°00'	112°45'	198°30'

- (b) The eye of an observer is 7 m above the sea level. He was able to see the top of a light-house 42 m high just at the

P.T.O.

level of the horizon. Determine the distance of the observer from the light house. [5]

3. (a) Explain the following terms : [6]

Vertical axis, Horizontal axis, Optical Plumet, Telescope normal.

- (b) Two tangents intersect at chainage of 1000 m with deflection angle of 55° . Calculate the chainage of T1 & T2, Length of long chord (L), Degree of curve (D), Apex distance, Mid ordinate distance, for setting out a curve with radius of 400 m. [6]

Or

4. (a) ABCDA is a closed traverse. Determine the missing data in the following table : [6]

Line	AB	BC	CD	DA
Length (m)	230.5	250.2	----	427
Bearing	N36°45'E	S82°48'E	----	N82°45'W

- (b) What are transition curves ? Give its necessity. [6]

5. (a) Enlist various permanent adjustments of a theodolite. Explain procedure to correct any *one*. [5]

- (b) A tacheometer with constants $K = 100$, $C = 0.3$ was used to observe the following readings : [8]

Instrument at	Staff at	Vertical angle	Staff Readings
	P	+ $5^\circ 15'$	1.355, 2.580, 3.935
A	Q	- $4^\circ 15'$	0.985, 1.660, 2.335

Determine the RL of Q. Take R.L. of P = 100.000 m. Also determine distance PQ if horizontal angle PAQ = $60^\circ 30'$.

Or

6. (a) Explain the procedure to find tacheometric constants on field. [4]

- (b) Derive the distance and elevation formulae for an inclined line of sight with angle of elevation and staff is vertical. [5]

(c)

Instrument at	Staff at	Distance(m)	Vertical Angle	Cross hair reading
P	A	80	2° 30'	1.325, 2.122
P	B	140	1° 36'	0.985, 2.382

Determine the techeometric constants from the given data.[4]

7. (a) What is total station ? What are the types of total station ? [6]

(b) Describe the method of laying alignment of drainage line.[7]

Or

8. (a) Explain the procedure for survey project on sanitary scheme.[6]

(b) Write a short note on the features of ETS. [7]